Lucas Y. Tian, Ph.D.

NINDS K99 Postdoctoral Associate
Laboratory of Neural Systems
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EDUCATION

2012 - 2019	Ph.D. in Neuroscience, UC San Francisco, CA (Advisor: Michael Brainard)
2007 - 2011	B.A. with Honors in Biology, specializing in Neuroscience, University of Chicago, IL

RESEARCH

2020 - Present	**Postdoctoral Scholar, The Rockefeller University**, New York, NY *Advisor: Winrich Freiwald; Co-advisors: Josh Tenenbaum, Xiao-Jing Wang* *Topic: Neural substrates of compositional generalization, in a macaque drawing task.*
2019 (May-Dec)	**Visiting Postdoctoral Scholar, MIT**, Cambridge, MA *Advisor: Josh Tenenbaum; Co-advisors: Winrich Freiwald, Xiao-Jing Wang* *Topic: Cognitive computational modeling of human drawing.*
2012 - 2019	**Graduate Student, UCSF**, San Francisco, CA *Advisor: Michael Brainard* *Thesis: Neural mechanisms of motor skill flexibility in songbirds (link)*

PEER-REVIEWED PUBLICATIONS

Tian LY, Warren TL, Mehaffey WH, Brainard MS (2023). Dynamic top-down biasing implements rapid adaptive changes to individual movements. *eLife* 12:e83223. Link

Veit L, **Tian LY**, Monroy Hernandez CJ, Brainard MS (2021). Songbirds can learn flexible contextual control over syllable sequencing. *eLife* 10: e61610. Link

Tian LY, Ellis K, Kryven M, Tenenbaum JB (2020). Learning abstract structure for drawing by efficient motor program induction. *Advances in Neural Information Processing Systems*, 33. Link

Tian LY, Brainard MS (2017). Discrete circuits support generalized versus context-specific vocal learning in the songbird. *Neuron* 96, 1-10. Link, Commentary, News

PREPRINTS (UNDER REVIEW)

Tian LY, Garzón KU, Hanuska DJ, Rouse AG, Eldridge MA, Schieber MH, Wang XJ, Tenenbaum JB, Freiwald WA (2025). Neural representation of action symbols in primate frontal cortex. *bioRxiv (*in revision at *Nature)*. 2025.03.03.641276. Link

Representative presentations: SfN 2024 poster (link), Cosyne 2025 talk (link).

MANUSCRIPTS IN PREPARATION

Tian LY, Hanuska DJ, Garzón KU, Liu Y, Tenenbaum JB, Wang XJ, Freiwald WA. Neural representation of an action grammar. *Expected submission Dec 2025.*

FUNDING

2023 - Present — **NINDS K99/R00 Transition to Independence Award**
Title: Investigating Symbolic Computation in the Brain: Neural Mechanisms of Compositionality (5K99NS131585) (PI: Tian, LY)
Total Award Amount (including Indirect Costs): $268,542 (2-year K99 phase), $747,000 (3-year R00 phase, estimated, if activated)

2021 - Present — **Simons Foundation SCGB Pilot Award** (*key personnel – contributing co-author*)
Title: Investigating Symbol-like Reasoning in the Brain: Neural Mechanisms of Compositional Action Planning (876120SPI) (PI: Freiwald, WA)
Total Award Amount (including Indirect Costs): $770,000

2020 - 2023 — **NIH BRAIN F32 NRSA Postdoctoral Fellowship**
Title: The planning of new compositional action sequences guided by interpretation of ambiguous sensory data in a novel drawing task (F32MH125573) (PI: Tian, LY)
*Total Award Amount: $*209,778

AWARDS AND HONORS

2025	MSN Extramural Postdoctoral Invited Speaker, Mt. Sinai
2025	SPiNES Extramural Postdoctoral Invited Speaker, NYU
2025	NEUROYES Extramural Postdoctoral Invited Speaker, University of Rochester
2023	NINDS K99/R00 Transition to Independence Award
2020	NIH BRAIN F32 NRSA Postdoctoral Fellowship
2020	LSRF Postdoctoral Fellowship Finalist
2014	Herbert W. Boyer PIBS Fellowship, UCSF
Travel grants	
2025	Postdoctoral Association Career Development Award, Rockefeller
2025	COSYNE Presenters Travel Grant
2016	Graduate Division Travel Award, UCSF
Undergraduate	
2011	Outstanding Thesis Award, UChicago (top two students in graduating class)
2011	Top Presentation in Biology, Chicago Area Undergraduate Research Symposium
2009	Amgen Scholarship, Caltech (David Anderson lab)
2007	National Merit College-sponsored Scholarship

SELECT CONFERENCE ABSTRACTS

Competitively selected talks	
2025	COSYNE Main Meeting (top 3% of submissions), Montreal, Canada (Recording)
2024	Ascona Circuits Meeting, Ascona, Switzerland (Slides)
2020	NeurIPS (top 6% of accepted papers) (Recording)
Talks	
2025	Society for Neuroscience Nanosymposium, San Diego, USA
Posters	
2025	Cognitive Computational Neuroscience, Amsterdam, The Netherlands
2025	Society for the Neural Control of Movement, Panama City, Panama

2024	Society for Neuroscience, Chicago, IL ([Link to poster])
2024	Ascona Circuits Meeting, Ascona, Switzerland

INVITED TALKS

Symposia

2025	MSN Extramural Postdoctoral Invited Speaker, Mt. Sinai, USA
2025	SPiNES Extramural Postdoctoral Invited Speaker, NYU, USA
2025	NEUROYES Extramural Postdoctoral Invited Speaker, University of Rochester, USA
2025	Kavli Internal Symposium, Rockefeller, New York, USA
2025	Lipschultz Symposium, Icahn School of Medicine (Mt. Sinai), New York, USA
2025	The Social Brain Symposium, Rockefeller and Columbia, New York, USA
2021	Object Cognition Workshop, Yale, New Haven, USA (virtual)
2018	Birds Bats Brains Meeting, UC Berkeley, USA

Departments

2025	Department of Psychology, UC San Diego, USA
2025	Department of Neurobiology, University of Pittsburgh, USA
2025	German Primate Center, Leibniz Institute for Primate Research, Göttingen, Germany
2023	Kavli Internal Seminar series, Rockefeller, New York, USA
2021	Center for Brains, Minds & Machines, MIT, Cambridge, USA ([Link to recording])
2021	PDA Summer Seminar Series, Rockefeller, New York, USA
2016	Asilomar Neuroscience Retreat, UC San Francisco, USA

Individual labs

2025	Stanford, Andreas Tolias (virtual)
2025	Western University, Jörn Diedrichsen, Paul Gribble, and Andrew Pruszynski (virtual)
2025	Princeton, Tim Buschman
2025	NYU, Marcelo Mattar
2025	MIT, Ev Fedorenko
2025	Newcastle University, Stuart Baker (virtual)
2025	NYU, Michael Long
2022	UCSF, Michael Brainard
2021	Yale, Ilker Yildirim (virtual)
2020	NYU, Brenden Lake and Todd Gureckis (virtual)
2020	MIT, Laura Schulz (virtual)
2016	UCSF, John Houde

SERVICE

2023 - Present	Peer-reviewed for: eLife, Cognitive Science Society, Cognitive Computational Neuroscience
2014 - 2019	Co-organizer, Systems Neuroscience Research in Progress Seminar Series, UCSF
2013 - 2019	Co-organizer, Special Topics Seminar Series, UCSF

MENTORSHIP

Research assistants

2024 - Present	**Daniel Hanuska**, research assistant, Rockefeller

2022 - 2024 **Kedar Garzón Gupta**, research assistant, Rockefeller
(Current position: Ph.D. Student at Columbia)

Ph.D. rotation students

2024 **Siddhartha Sharma**, Freiwald lab, Rockefeller
2021 **Yanis Tazi**, Freiwald lab, Rockefeller
2018 **Eszter Kish**, Brainard lab, UCSF
2014 **Rachel Care**, Brainard lab, UCSF

Interns and other mentorship

2024 - 2025 **Valerie Calligy**, Shenoy Undergraduate Research Fellow (SURFiN), Simons Foundation, Rockefeller
2024 - 2025 **Xuan Ma**, Bard-Rockefeller intern, Rockefeller
2024 **Daniel Dolnik**, intern, Cornell Tech Master's Program
2024 **Thea Wu**, summer intern (Summer Undergraduate Research Fellowship, SURF), Rockefeller
2022 **Alejandra Urquieta**, summer intern, SURF, Rockefeller
2021 **Sam Coolsaet**, medical intern, Rockefeller
2021 **Alex Gu**, summer intern, SURF, MIT
2017 **Christian Jose Monroy Hernandez**, HHMI Exceptional Research Opportunities Program (EXROP), UCSF (co-mentor: Lena Veit)

OUTREACH

2024 - 2025 Mentor, Simons Foundation SURFiN, Rockefeller
Mentee: Valerie Calligy
2021 Mentor, Neuromatch Academy (virtual)
Mentees: Pulido Arias, Lourdes Baztan Bitopoulos, Joseph Gonzalez, May Xia, Mark Zarutin
2017 Mentor, HHMI EXROP, UCSF
Mentee: Christian Jose Monroy Hernandez (co-mentor: Lena Veit)
2014 Judge, Synopsys Science and Technology Championship, San Jose, CA
2013 Invited speaker, PITCH, Center for Education Partnerships (CEP), UCSF
2013 - 2014 Mentor, Fairposium at Burton High School, CEP, UCSF

TEACHING

2014 TA, BMS117: Infection & Host Response (Neurophysiology unit), UCSF
2012 - 2013 Scientist-Teacher, Raul Wallenberg High School, Science and Health Education Partnership, UCSF
2012 English Teacher, San Gabriel Kindergarten, Los Nogales, Peru
2010 Teacher, Neuroscience of Illusions, Cascade!, University of Chicago

ADVANCED COURSES

2019 Methods in Computational Neuroscience, Marine Biological Laboratory, MA
2018 Jr. Workshop on Mechanistic Cognitive Neuroscience, Janelia Research Campus, VA
2018 Brains, Minds and Machines, Marine Biological Laboratory, MA
2017 Mining and Modeling of Neuroscience Data, Redwood Center, UC Berkeley, CA

REFERENCES

Winrich Freiwald, Ph.D.
Denise A. and Eugene W. Chinery Professor
Laboratory of Neural Systems, Rockefeller University
1230 York Avenue, New York, NY 10065
wfreiwald@rockefeller.edu
(212) 327-8000
Relationship: Postdoctoral advisor

Michael Brainard, Ph.D.
Professor of Physiology and Psychiatry
Departments of Physiology and Psychiatry, UCSF/HHMI
675 Nelson Rising Lane, Room 514E, San Francisco, CA 94158
michael.brainard@ucsf.edu
(415) 502-7344
Relationship: Ph.D. advisor

Joshua Tenenbaum, Ph.D.
Professor of Computational Cognitive Science
Department of Brain and Cognitive Sciences, MIT
Building 46-4015, 77 Massachusetts Avenue, Cambridge, MA 02139
jbt@mit.edu
(617) 452-2010
Relationship: Postdoctoral co-advisor

Xiao-Jing Wang, Ph.D.
Professor of Neural Science
Center for Neural Science, NYU
4 Washington Place, Room 752, New York, NY 10003
xjwang@nyu.edu
(212) 998-3677
Relationship: Postdoctoral co-advisor